

Problem 7.1

$$a) A = \begin{bmatrix} 1 & 0 & 2\frac{3}{4} & 2\frac{9}{4} \\ 0 & 1 & \frac{1}{4} & \frac{7}{4} \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

b) 2

$$c) \begin{bmatrix} -2\frac{3}{4} \\ -\frac{1}{4} \\ 1 \\ 0 \end{bmatrix}$$

Problem 7.2

$$A_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A_2 = \begin{bmatrix} 1 & -1 \\ -1 & -1 \end{bmatrix}$$